Machine Learning and Artificial Intelligence

Artificial Intelligence is a field of computer science focused on creating systems capable of performing tasks that normally require human intelligence. These tasks can include understanding language, recognizing images, making predictions, solving problems, and generating content.

Machine Learning is a branch of artificial intelligence that allows computers to learn patterns from data. Instead of manually programming every decision, developers train a machine learning model using existing data so that it can make predictions about new information.

Supervised learning uses labeled data where the expected output is already known. Two major supervised learning problems are regression and classification. Regression predicts numerical values such as house prices, while classification predicts categories such as whether an email is spam or not spam.

A machine learning dataset usually contains features and a target. Features represent the information given to the model, while the target represents what the model is expected to predict. The dataset is commonly divided into training and testing data so that the model can be evaluated using information it has not previously seen.

Machine learning models are evaluated using different metrics. Classification models commonly use accuracy, precision, recall, F1-score, and confusion matrices. Regression models commonly use metrics such as Mean Absolute Error, Mean Squared Error, Root Mean Squared Error, and R-squared.

Deep learning is a branch of machine learning based on neural networks containing multiple layers. Deep learning is widely used in computer vision, speech recognition, natural language processing, and generative artificial intelligence.

Transformers are neural network architectures that use attention mechanisms to understand relationships between pieces of information. Modern Large Language Models are commonly based on transformer architectures and can perform tasks such as question answering, summarization, translation, text generation, and code generation.